

Independent technical study — companion report to the simulation code

Field-Oriented Control of a Squirrel-Cage Induction Motor

Modelling, vector-control design and three-way simulation validation

Electrical drives and power electronics

A 5 kW induction motor drive: indirect rotor-flux orientation, cascaded PI control and four-quadrant operation

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Abstract

This report presents the design, implementation and validation of an indirect field-oriented control (IFOC) scheme for a 5 kW, 380 V squirrel-cage induction motor. The machine is represented by its two-axis state-space model written in the stationary $\alpha\beta$ frame, with stator currents and rotor flux linkages as state variables. The controller performs rotor-flux orientation using a slip-based angle generator and a current-model flux estimator, and regulates the machine through two nested control levels: an outer speed loop and two inner current loops equipped with cross-coupling compensation, back-calculation anti-windup and explicit torque, current and voltage limiting.

Two complementary test scenarios are used. The first applies a staircase of constant speed references (800, 1400 and 600 rpm followed by a reversal to -800 rpm) with a 10 N·m load-torque disturbance, and exercises the drive as a machine: every step settles within 71–95 ms without overshoot, the residual speed error on the settled plateaus is 0.05 rpm RMS, the load step produces a 69 rpm dip recovered in about 70 ms, and the rotor flux remains locked on its 0.8 Wb reference with a quadrature residual below 2 %. The second scenario applies ramp and sinusoidal references to the unloaded machine and exercises the speed loop as a servo; the measured errors confirm the internal-model prediction quantitatively, with zero steady-state error on the ramp and a frequency-dependent sinusoidal error matching the analytical estimate within roughly 9 %.

The drive was then rebuilt twice in MATLAB/Simulink, once from fundamental blocks with an average-value inverter and once from Simscape Electrical components with a true 10 kHz switching bridge. The three independent implementations agree on every physically meaningful quantity, which provides a strong cross-validation of the machine model, the control design and the parameter set.

Keywords: induction machine, vector control, indirect field orientation, Clarke and Park transformations, space-vector modulation, cascaded PI control, anti-windup, four-quadrant drive, Simscape Electrical.

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Nomenclature

$v_{s\alpha}, v_{s\beta}$	Stator voltage components, stationary frame	V
$i_{s\alpha}, i_{s\beta}$	Stator current components, stationary frame	A
$\psi_{r\alpha}, \psi_{r\beta}$	Rotor flux linkage components, stationary frame	Wb
v_{sd}, v_{sq}	Stator voltage components, rotor-flux frame	V
i_{sd}, i_{sq}	Flux- and torque-producing current components	A
ψ_r	Rotor flux linkage magnitude	Wb
θ_e	Rotor flux angle (electrical)	rad
ω_m	Mechanical rotor speed	rad/s
ω_r	Electrical rotor speed, $\omega_r = p \omega_m$	rad/s
ω_{sl}	Slip angular frequency	rad/s
ω_e	Synchronous (flux) angular frequency	rad/s
T_e, T_L	Electromagnetic and load torque	N·m
R_s, R_r	Stator and referred rotor resistance	Ω
L_s, L_r, L_m	Stator, rotor and magnetising inductance	H
σ	Total leakage coefficient, $1 - L_m^2/(L_s L_r)$	–
τ_s, τ_r	Stator and rotor time constant	s
R_σ	Equivalent transient resistance	Ω
p	Number of pole pairs	–
J, f_r	Rotor inertia and viscous friction coefficient	kg·m ² , N·m·s
V_{dc}	Inverter DC-link voltage	V
T_s	Controller sampling period	s
K_p, K_i	Proportional and integral controller gains	–
ω_{ci}, ω_{cs}	Current- and speed-loop crossover frequencies	rad/s

Acronyms. FOC: field-oriented control. IFOC: indirect field-oriented control. VSI: voltage-source inverter. SVPWM: space-vector pulse width modulation. PWM: pulse width modulation. SPS: Specialized Power Systems (Simscape Electrical). RK4: fourth-order Runge–Kutta. RMS: root mean square.

1 Introduction

1.1 Context and motivation

The three-phase squirrel-cage induction machine remains the most widely used electromechanical converter in industry. It has no brushes, no permanent magnets and no rotor winding to connect, which makes it cheap, mechanically robust and almost maintenance-free. These advantages come at a price on the control side. In a separately excited DC machine the field flux and the armature current are carried by two physically distinct circuits that are naturally orthogonal, so torque can be commanded directly and instantaneously through the armature current. In an induction machine there is no such separation: torque is produced by the interaction of two quantities that are both alternating, mutually coupled and buried in a set of nonlinear differential equations.

Scalar control strategies, in particular constant V/f operation, sidestep the problem by regulating only the magnitude and frequency of the stator voltage. They are simple and adequate for pumps and fans, but they control the flux only in the average sense and leave the torque dynamics slow and poorly damped. Any application that requires fast torque response, accurate speed regulation at low speed, or controlled operation through zero speed needs a different approach.

Field-oriented control, proposed by Blaschke [1] and Hasse [2] in the early 1970s, provides that approach. Its central idea is a change of viewpoint: instead of controlling the machine in a frame attached to the stator, the machine is controlled in a frame attached to the rotor flux. In that particular frame, and only there, the induction machine behaves like a separately excited DC machine. One current component sets the flux, the other sets the torque, and both become constant quantities in steady state, so ordinary PI controllers regulate them with zero steady-state error. The price to pay is that the frame must be located accurately, which is what most of the control design is actually about.

1.2 Objectives of the study

The work reported here builds a complete field-oriented drive for a 5 kW induction machine and validates it in simulation. The objectives were set as follows.

1. Derive and implement a dynamic model of the machine suitable for control design, and state clearly which assumptions it relies on.
2. Implement the coordinate transformations and the indirect orientation mechanism, including the flux estimator, rather than assuming the flux angle is known.
3. Design the cascaded controllers analytically, from the machine parameters, and justify every gain and every limit.
4. Validate the drive on two deliberately different test scenarios: one that exercises the machine at its limits, and one that isolates the linear behaviour of the speed loop.
5. Reproduce the whole drive independently on a second platform, including a model with a genuine switching inverter, and compare the results quantitatively.

The fifth objective deserves a comment. A single simulation proves very little: a model and its controller can be wrong in a mutually consistent way and still produce a plausible plot. Building the same drive three times, with two different tools and two different inverter representations, turns the exercise into a genuine cross-check. Where the three implementations agree, the result is attributable to the physics; where they differ, the difference must

be explained by what each model actually represents. That discipline is applied throughout Section 8.

1.3 Outline

Section 2 derives the machine model and lists the parameter set. Section 3 presents the Clarke and Park transformations and explains why they are applied in two stages. Section 4 develops the field-oriented control structure, the flux estimator, the decoupling terms, the limiting strategy and the analytical tuning. Section 5 describes the simulation environment and the two test scenarios. Sections 6 and 7 present and discuss the results of each scenario. Section 8 covers the two Simulink implementations and the three-way comparison. Section 9 discusses the limitations of the study and Section 10 concludes.

2 Modelling of the induction machine

2.1 Assumptions

The model used throughout this report is the classical two-axis dynamic model. It rests on the following assumptions, which are standard for drive-level studies and are stated explicitly because several of them reappear in the discussion of Section 9.

- The three stator windings are identical, sinusoidally distributed and displaced by 120° in space; the squirrel cage is represented by an equivalent three-phase winding referred to the stator.
- The magnetic circuit is linear: saturation and hysteresis are neglected, so all inductances are constant.
- Iron losses are neglected. Only the copper losses appear, through R_s and R_r .
- Slotting effects and space harmonics are ignored, so the air-gap flux density is taken as sinusoidally distributed.
- The machine is symmetrical and the neutral is isolated, hence $i_a + i_b + i_c = 0$ and the zero-sequence component vanishes.
- Parameters are constant. In particular the rotor resistance R_r does not vary with temperature, an assumption revisited in Section 9.

2.2 Machine data

The machine studied is a 5 kW, 380 V, four-pole squirrel-cage motor. Its parameters are given in Table 2; they correspond to a standard laboratory machine and are the values used by all three implementations of the drive.

Four derived quantities appear repeatedly in the equations and in the controller design, and are worth computing once and for all:

$$\sigma = 1 - \frac{L_m^2}{L_s L_r} = 0.0555, \quad \tau_r = \frac{L_r}{R_r} = 87.0 \text{ ms}, \quad (1)$$

$$\tau_s = \frac{L_s}{R_s} = 163.2 \text{ ms}, \quad \sigma L_s = 3.94 \text{ mH}. \quad (2)$$

The leakage coefficient σ is small, as expected for a machine with a narrow air gap: only 5.5% of the stator flux fails to link the rotor. The product σL_s is the transient inductance, that is, the inductance seen by a fast current change; it is roughly eighteen times smaller than L_s ,

Table 2. Rated data and equivalent-circuit parameters of the induction machine.

Quantity	Symbol	Value	Unit
Rated power	P_n	5000	W
Rated line-to-line voltage (RMS)	U_n	380	V
Rated current	I_n	10.5	A
Rated frequency	f_n	50	Hz
Pole pairs	p	2	–
Stator resistance	R_s	0.435	Ω
Rotor resistance (referred)	R_r	0.816	Ω
Stator inductance (total)	L_s	0.071	H
Rotor inductance (total)	L_r	0.071	H
Magnetising inductance	L_m	0.069	H
Rotor inertia	J	0.0056	$\text{kg}\cdot\text{m}^2$
Viscous friction	f_r	0.0015	$\text{N}\cdot\text{m}\cdot\text{s}$

and it is this small value, not L_s , that determines how fast the current loops can be made. The rotor time constant τ_r sets the rate at which the flux can be built up or changed, and reappears in the slip relation of Section 4.2.

2.3 State-space model in the stationary frame

Several equivalent formulations of the induction machine model exist; they differ by the choice of reference frame and of state variables. The formulation adopted here uses the stationary $\alpha\beta$ frame with the stator currents and the rotor flux linkages as states:

$$\mathbf{x} = [i_{s\alpha} \quad i_{s\beta} \quad \psi_{r\alpha} \quad \psi_{r\beta}]^T. \quad (3)$$

Starting from the stator and rotor voltage equations written in the stationary frame and eliminating the rotor currents through the flux definition $\vec{\psi}_r = L_r \vec{i}_r + L_m \vec{i}_s$, the model takes the form

$$\frac{d\psi_{r\alpha}}{dt} = -\frac{\psi_{r\alpha}}{\tau_r} - \omega_r \psi_{r\beta} + \frac{L_m}{\tau_r} i_{s\alpha}, \quad (4)$$

$$\frac{d\psi_{r\beta}}{dt} = -\frac{\psi_{r\beta}}{\tau_r} + \omega_r \psi_{r\alpha} + \frac{L_m}{\tau_r} i_{s\beta}, \quad (5)$$

$$\frac{di_{s\alpha}}{dt} = \frac{1}{\sigma L_s} \left[v_{s\alpha} - R_\sigma i_{s\alpha} + \frac{L_m}{L_r \tau_r} \psi_{r\alpha} + \frac{L_m}{L_r} \omega_r \psi_{r\beta} \right], \quad (6)$$

$$\frac{di_{s\beta}}{dt} = \frac{1}{\sigma L_s} \left[v_{s\beta} - R_\sigma i_{s\beta} + \frac{L_m}{L_r \tau_r} \psi_{r\beta} - \frac{L_m}{L_r} \omega_r \psi_{r\alpha} \right], \quad (7)$$

where $\omega_r = p \omega_m$ is the electrical rotor speed and

$$R_\sigma = R_s + R_r \left(\frac{L_m}{L_r} \right)^2 = 1.206 \, \Omega \quad (8)$$

is the equivalent resistance seen by the transient current dynamics.

The electromagnetic torque follows from the cross product of the rotor flux and the stator current vectors. With the amplitude-invariant convention adopted here (see Section 3.2), a factor $3/2$ appears:

$$T_e = \frac{3}{2} p \frac{L_m}{L_r} (\psi_{r\alpha} i_{s\beta} - \psi_{r\beta} i_{s\alpha}), \quad (9)$$

and the mechanical equation closes the model:

$$J \frac{d\omega_m}{dt} = T_e - T_L - f_r \omega_m. \quad (10)$$

2.4 Why this particular formulation

An equally common formulation keeps the four currents $[i_{s\alpha}, i_{s\beta}, i_{r\alpha}, i_{r\beta}]^T$ as states and writes the model as $\mathbf{L} \dot{\mathbf{x}} = -(\mathbf{R} + \omega_r \mathbf{A})\mathbf{x} + \mathbf{B}\mathbf{v}$. The two descriptions are strictly equivalent, being related by the linear change of variables $\vec{\psi}_r = L_r \vec{i}_r + L_m \vec{i}_s$. The flux-based form was preferred here for two practical reasons.

First, the rotor flux is exactly the physical quantity on which the controller orients itself, so having it as a state makes the orientation error directly observable: the controller estimates the flux independently, and the difference between the estimate and the model state is available at every instant. Had the controller been given the true flux angle, field orientation would have been assumed rather than demonstrated, and the results of Section 6.1 would carry no information.

Second, the flux states are slow. Their time constant is $\tau_r = 87$ ms, whereas the current states evolve with the transient time constant $\sigma L_s / R_\sigma \approx 3.3$ ms. Keeping the slow variables explicit improves the conditioning of the integration and allows a comfortable fixed integration step.

3 Reference-frame transformations

3.1 The two problems to be solved

The stator carries three currents displaced by 120° . Two separate difficulties follow from this, and it is useful to recognise them as distinct because each transformation solves exactly one of them.

The first difficulty is redundancy. With an isolated neutral, $i_a + i_b + i_c = 0$, so only two of the three phase quantities are independent. Manipulating three coupled variables when two suffice complicates every subsequent step without adding information.

The second difficulty is that, in any frame fixed to the stator, all electrical quantities are sinusoidal at the supply frequency. A PI controller applied to a sinusoidal reference cannot achieve zero steady-state error, because its open-loop transfer function contains no resonant term at the excitation frequency. Section 7.1 returns to this point in detail; for now it is enough to note that regulating sinusoids with PI controllers is structurally unsatisfactory.

3.2 Clarke transformation: removing the redundancy

The Clarke transformation projects the three phase axes onto two orthogonal stationary axes: α aligned with the axis of phase a , and β in quadrature with it. The zero-sequence component

is discarded. In the amplitude-invariant form used here,

$$\begin{bmatrix} x_\alpha \\ x_\beta \end{bmatrix} = \frac{2}{3} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} x_a \\ x_b \\ x_c \end{bmatrix}. \quad (11)$$

The factor $2/3$ is a convention, not a physical requirement. It is chosen so that the magnitude of the resulting space vector equals the peak value of the phase quantities, which makes limits and set-points directly interpretable: a current limit of 25 A in the $\alpha\beta$ plane really is a peak phase current of 25 A. The consequence of this choice is the factor $3/2$ in the torque expression (9); had the power-invariant scaling $\sqrt{2/3}$ been used instead, the torque expression would carry no numerical factor but the current magnitudes would no longer match the phase peaks. Both conventions are correct provided they are used consistently, and mixing them is a classical source of a factor $\sqrt{3/2}$ in torque.

After the Clarke transformation the machine is described as an equivalent two-phase machine. The quantities are still alternating: the current space vector $\vec{i}_s = i_{s\alpha} + j i_{s\beta}$ rotates at the supply frequency. Only the redundancy has been removed.

3.3 Park transformation: stopping the rotation

The Park transformation re-expresses the $\alpha\beta$ vector in a frame rotating at an angle θ , which for field-oriented control is the rotor-flux angle θ_e :

$$\begin{bmatrix} x_d \\ x_q \end{bmatrix} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x_\alpha \\ x_\beta \end{bmatrix}. \quad (12)$$

Seen from a frame that rotates with the flux vector, steady-state sinusoids become constants. This is the decisive step of the whole method. Three consequences follow immediately: PI controllers acting on dq quantities have zero steady-state error, feed-forward terms and saturation limits become meaningful constant values, and the flux/torque structure of the machine becomes visible, as shown in Section 4.1.

3.4 Why the transformation is applied in two stages

Conceptually, Clarke followed by Park is a single transformation from abc to dq , and it could be implemented as one 2×3 matrix with angle-dependent coefficients. Splitting it is nonetheless standard practice, for three reasons.

Separation of concerns. Clarke deals with the geometry of the winding: three axes become two orthogonal axes and the zero sequence is removed. Park deals with the dynamics: a stationary description becomes a rotating one. Keeping them separate isolates a constant, purely algebraic step from the angle-dependent step.

Computational cost. The Clarke matrix has constant coefficients and requires no trigonometric evaluation. Only Park needs $\sin \theta_e$ and $\cos \theta_e$, and these two values are computed once per sampling period and shared by the measurement path ($abc \rightarrow dq$) and the actuation path ($dq \rightarrow abc$). On a fixed-point DSP this matters.

The intermediate signals are useful. The $\alpha\beta$ quantities are exactly what the space-vector modulator consumes (Section 4.6) and what stationary-frame observers use in sensorless drives.

The chain $abc \rightarrow \alpha\beta \rightarrow dq$ exposes them at no extra cost.

The inverse path applies the inverse Park transformation to the controller's voltage commands v_d^*, v_q^* and then the inverse Clarke transformation, producing the three-phase voltage references that the modulator must synthesise.

4 Field-oriented control

4.1 Principle of rotor-flux orientation

Choose the d axis of the Park frame to lie along the rotor flux vector. By construction $\psi_{rq} = 0$ and $\psi_{rd} = \psi_r$. Substituting this into the machine equations written in the rotating frame collapses them to

$$\tau_r \frac{d\psi_r}{dt} + \psi_r = L_m i_{sd}, \quad (13)$$

$$T_e = \frac{3}{2} p \frac{L_m}{L_r} \psi_r i_{sq}. \quad (14)$$

Equation (13) states that the rotor flux is governed by i_{sd} alone, through a first-order lag of time constant τ_r . Equation (14) states that, at constant flux, the torque is proportional to i_{sq} . The analogy with the separately excited DC machine is now complete: i_{sd} plays the role of the field current and i_{sq} that of the armature current, and the two channels are independent.

This result is conditional. It holds if and only if the frame is genuinely aligned with the rotor flux, that is, if θ_e is known accurately. Locating that angle is therefore the real problem, and it is the subject of the next subsection.

4.2 Indirect orientation: locating the flux without a sensor

The rotor flux of a squirrel-cage machine is not directly measurable: there is no accessible rotor winding and no flux sensor in a standard motor. Two families of solutions exist. Direct field orientation estimates the flux vector from measured terminal quantities using an observer. Indirect field orientation, due to Hasse [2], computes the angle from the slip relation which holds exactly when the frame is oriented:

$$\omega_{sl} = \frac{L_m i_{sq}}{\tau_r \psi_r}, \quad \theta_e = \int (p \omega_m + \omega_{sl}) dt. \quad (15)$$

The angle is thus obtained by adding a computed slip to the measured rotor position and integrating. The flux magnitude appearing in (15) is provided by a current-model estimator, which is nothing but the rotor lag of (13) run as an observer:

$$\hat{\psi}_r = \frac{L_m}{1 + s \tau_r} i_{sd}. \quad (16)$$

Using the estimate rather than the reference value matters during flux build-up. At start-up the true flux is zero while its reference is 0.8 Wb; feeding the reference into (15) would produce a slip that is far too small and a torque scaling that is far too optimistic, and the drive would lose orientation precisely when it is most fragile. Feeding the estimate keeps both quantities consistent with the actual state of the machine.

The scheme has a well-known weakness which should be stated at the outset: both (15) and (16) contain $\tau_r = L_r/R_r$, and R_r varies by up to 50% between a cold and a hot rotor. A detuned τ_r introduces a steady orientation error, which degrades the decoupling and the torque accuracy without producing any instability, so it is easy to miss. Section 9 returns to this point.

4.3 Control architecture

The complete structure is shown in Figure 1. It is organised as a cascade with two levels.

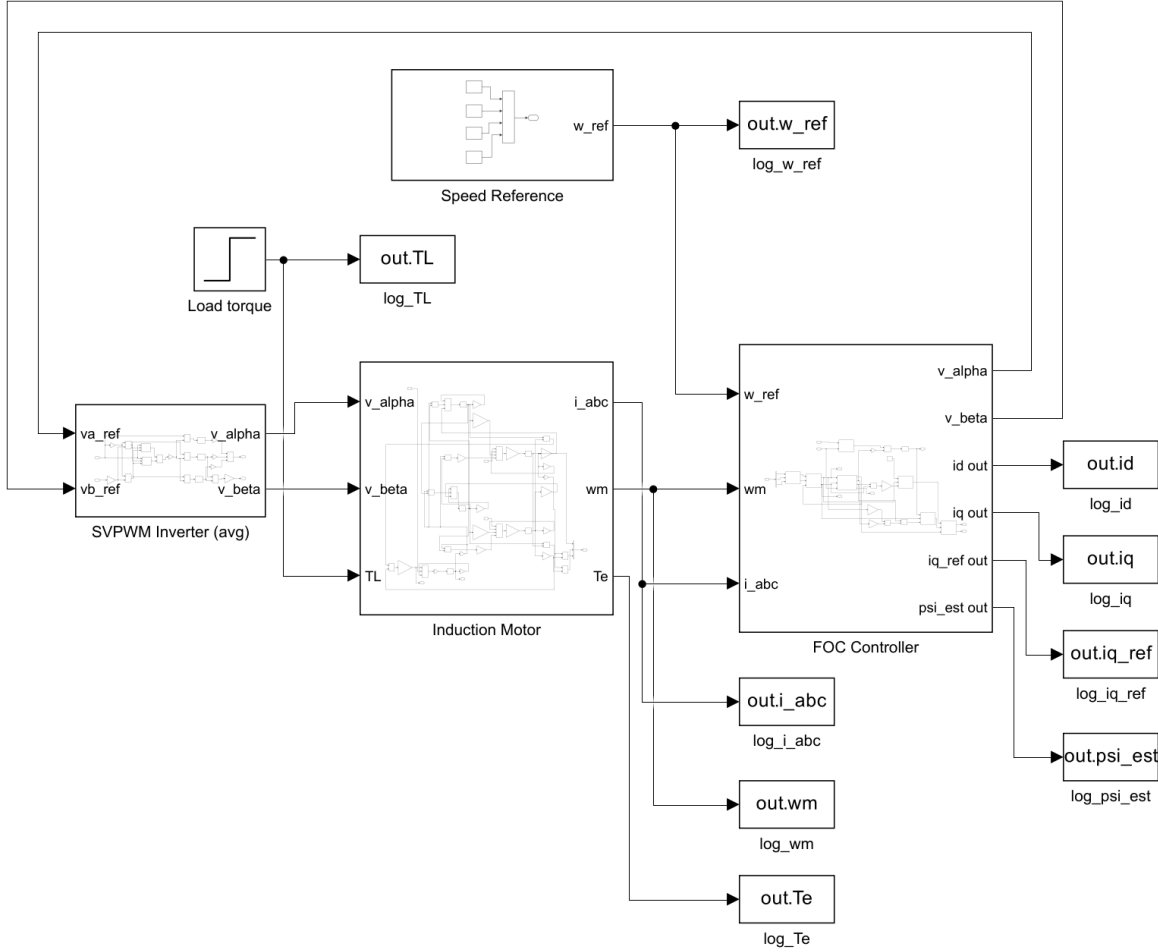
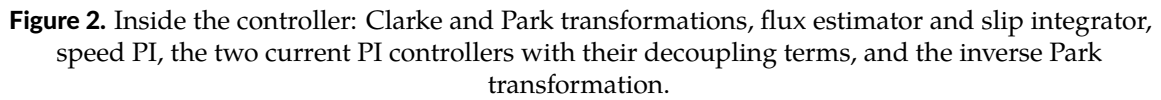


Figure 1. Top-level structure of the drive, as implemented in the fundamental-block Simulink model: speed reference, field-oriented controller, inverter and machine, with current and speed feedback.

The outer level is the speed loop. Its PI controller converts the speed error into a torque demand T_e^* , which is then divided by $\frac{3}{2}p\frac{L_m}{L_r}\hat{\psi}_r$ to produce the torque-current reference i_{sq}^* , in accordance with (14). The flux channel needs no controller in the constant-flux region: since the rotor lag (13) is already a stable first-order system with unity DC gain when written in terms of $L_m\dot{i}_{sd}$, the reference $i_{sd}^* = \psi_r^*/L_m = 11.6$ A is applied directly. A flux PI controller would be required only if field weakening were implemented.

The inner level consists of two current controllers, one per axis, sampled at 10 kHz. They act on dq quantities, so their references are constant in steady state and they achieve zero steady-state error. Their outputs are the voltage commands v_d^* and v_q^* .



The two current channels are not naturally independent. Writing the stator equations in the rotating frame produces cross terms proportional to ω_e which couple the d and q axes and act as disturbances on each other. They are cancelled by feed-forward:

$$v_{q,\text{ff}} = +\omega_e \left(\sigma L_s i_{sd} + \frac{L_m}{L_r} \hat{\psi}_r \right). \quad (18)$$

$$G_i(s) = \frac{1}{\sigma L_s s + R_\sigma}. \quad (19)$$

A drive without limits is not a drive. Three limits are enforced, in a definite order of priority.

- **Torque limit.** The output of the speed PI is clamped to ± 35 N·m, roughly 2.2 times the rated torque. This limits the acceleration and, indirectly, the current.
- **Current limit.** The stator current vector is limited to 25 A peak. The d axis has priority, because losing flux is worse than losing torque: the q -axis reference is limited to $i_{q,\max} = \sqrt{I_{\max}^2 - i_{sd}^{*2}} = 22.1$ A.
- **Voltage limit.** The voltage vector is limited to the linear SVPWM range, again with d -axis priority.

Every PI controller in the drive carries back-calculation anti-windup: the difference between the saturated and unsaturated outputs is fed back into the integrator through a gain K_b . This is not a cosmetic refinement. Section 8.1 reports the measured effect: with plain integrator clamping, the speed loop overshoots to about 960 rpm on an 800 rpm step, because the integral state charges up during the saturated phase and must then be discharged. With back-calculation the overshoot disappears entirely.

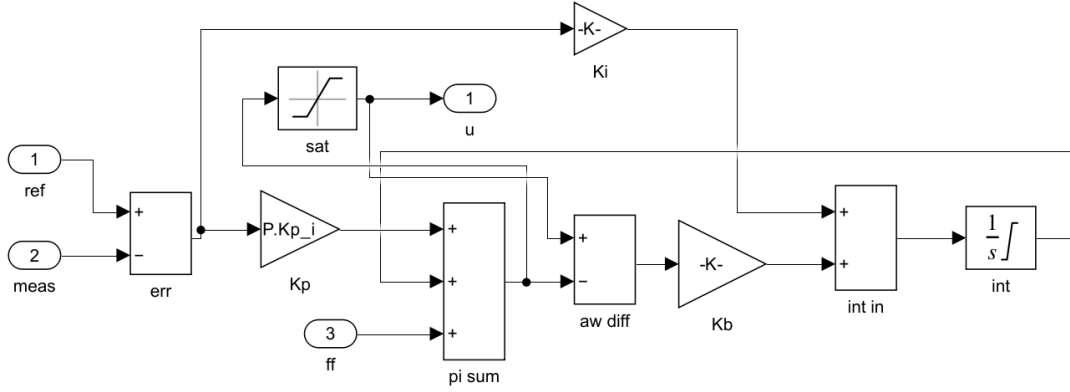


Figure 3. Structure of each PI controller: proportional and integral paths, output saturation, and the back-calculation anti-windup loop feeding the saturation error back into the integrator through K_b .

4.6 Inverter and space-vector modulation

The machine is fed by a two-level voltage-source inverter supplied from a DC bus of $V_{dc} = 540$ V, which corresponds to the natural output of a diode rectifier fed from the 380 V mains ($\sqrt{2} \times 380 \approx 537$ V).

Rather than implementing the sector-and-timing algorithm explicitly, the modulator uses the equivalent min–max common-mode injection:

$$v_{cm} = -\frac{1}{2} \left(\max(v_a^*, v_b^*, v_c^*) + \min(v_a^*, v_b^*, v_c^*) \right), \quad (20)$$

which is added to the three phase references before comparison with the carrier. This produces exactly the same duty cycles as classical space-vector modulation, at a fraction of the computational cost, and extends the linear modulation range to a phase voltage amplitude of

$$\hat{V}_{\max} = \frac{V_{dc}}{\sqrt{3}} = 311.8 \text{ V}, \quad (21)$$

which is 15.5 % higher than the $V_{dc}/2 = 270$ V achievable with sinusoidal PWM. The whole operating envelope of the study was chosen to remain inside this limit, so that field weakening is never required and the constant-flux assumption of Section 4.1 holds throughout.

For the Python and the fundamental-block Simulink models the inverter is represented by its cycle-averaged model, that is, the commanded voltage is applied directly. The switching model with a real bridge is used in the Simscape Electrical implementation of Section 8.2, and the comparison between the two quantifies exactly what averaging discards.

4.7 Controller tuning

With the decoupling active, each current loop sees the first-order plant (19). Choosing the PI zero to cancel the plant pole, $K_p/K_i = \sigma L_s/R_\sigma$, reduces the open loop to a pure integrator whose crossover frequency is set directly by the gains:

$$K_{p,i} = \sigma L_s \omega_{ci}, \quad K_{i,i} = R_\sigma \omega_{ci}. \quad (22)$$

For the speed loop the mechanical plant is $1/(Js + f_r)$, and since f_r is very small the plant is essentially a pure integrator. The gains are set from the desired crossover ω_{cs} :

$$K_{p,s} = J \omega_{cs}, \quad K_{i,s} = \frac{K_{p,s} \omega_{cs}}{5}, \quad (23)$$

where the factor 5 places the PI zero one fifth of a decade below crossover, a standard compromise giving a phase margin close to 60° .

The numerical values are collected in Table 3.

Table 3. Controller design rules, chosen bandwidths and resulting gains.

Loop	Design rule	Bandwidth	K_p	K_i
Current d, q	pole-zero cancellation	400 Hz	9.9 V/A	3030 V/(A·s)
Speed	integrator + zero at $\omega_{cs}/5$	30 Hz	1.06 N·m·s	39.8 N·m

Two design choices deserve justification.

Why 400 Hz for the current loops. The upper bound is set by the sampling frequency. With $T_s = 100 \mu\text{s}$ the sampling frequency is 10 kHz, and a widely used rule of thumb keeps the closed-loop bandwidth below one twentieth of the sampling frequency to leave the discretisation and the modulator delay outside the loop bandwidth. 400 Hz is exactly one twenty-fifth, which is comfortable.

Why 30 Hz for the speed loop. The lower bound comes from the disturbance-rejection requirement rather than from tracking. For a step load torque ΔT_L applied to an inertia J with a speed loop of crossover ω_{cs} , the transient speed dip scales approximately as $\Delta\omega \approx \Delta T_L / (J \omega_{cs})$. With $J = 0.0056 \text{ kg}\cdot\text{m}^2$ and $\Delta T_L = 10 \text{ N}\cdot\text{m}$, a 15 Hz bandwidth would give a dip of roughly 180 rpm, which is unacceptable for the intended application. Doubling the bandwidth to 30 Hz halves the dip; the value measured in simulation is 69 rpm (Section 6.2), in line with the estimate. The upper bound is the usual cascade requirement that the outer loop be significantly slower than the inner one; the ratio here is $400/30 \approx 13$, which is ample.

5 Simulation environment and test scenarios

5.1 Numerical implementation

The plant is integrated with a fixed-step fourth-order Runge–Kutta scheme at $20 \mu\text{s}$, that is, five integration sub-steps per control period. The choice of a fixed step, rather than a variable-step solver, was deliberate: it makes the simulation deterministic and directly comparable with the discrete Simulink models.

The controller is implemented as a genuinely discrete system running at $T_s = 100 \mu\text{s}$. Between two sampling instants no controller state is updated: the integrators hold, the transformations are not recomputed, and the voltage command applied to the inverter is the one computed at the last sampling instant. This mirrors what happens in a real DSP implementation, and it is the reason the Python results show slightly slower settling times than the continuous-controller Simulink model of Section 8.1. Measurements of current and speed are assumed ideal, with no noise, quantisation or delay.

5.2 Scenario A: speed staircase with load disturbance

The first scenario exercises the drive as a machine, deliberately driving it into its torque limit and through zero speed. Its profile is given in Table 4.

Table 4. Scenario A: staircase of constant speed references with a load step.

Instant / interval	Reference	Purpose
0 – 0.30 s	0 rpm	flux build-up, magnetisation to 0.8 Wb
0.30 s	step 0 → 800 rpm	acceleration from standstill
0.55 s	—	load step of 10 N·m, held to the end
0.80 s	step 800 → 1400 rpm	acceleration under load
1.30 s	step 1400 → 600 rpm	deceleration, regenerative braking
1.80 s	step 600 → –800 rpm	four-quadrant reversal through zero
1.80 – 2.40 s	hold –800 rpm	steady reversed operation under load

Constant set-points reached through steps were chosen deliberately as the test class. As Section 7.1 shows, a PI-based cascade tracks this class of reference with exactly zero steady-state error. Any deviation observed in the results is therefore attributable to the physics of the drive, such as torque limiting or load disturbances, and not to a structural limitation of the controller. This separation of causes is what makes the results interpretable.

5.3 Scenario B: ramp and sinusoidal tracking

The second scenario exercises the speed loop as a servo. It is run without load and with references gentle enough that no limit is ever reached, so that the measured errors reflect the linear behaviour of the loop alone.

Table 5. Scenario B: reference profile for the tracking study.

Interval	Reference	Purpose
0 – 0.30 s	0 rpm	flux build-up
0.30 – 0.90 s	ramp 0 → 1000 rpm	ramp tracking, type-2 behaviour
0.90 – 1.20 s	hold 1000 rpm	settling
1.20 – 1.70 s	sine 1000 ± 150 rpm at 2 Hz	slow sinusoidal tracking
1.70 – 2.30 s	sine 1000 ± 150 rpm at 5 Hz	faster sinusoidal tracking

6 Results: scenario A

6.1 Rotor flux and quality of the field orientation

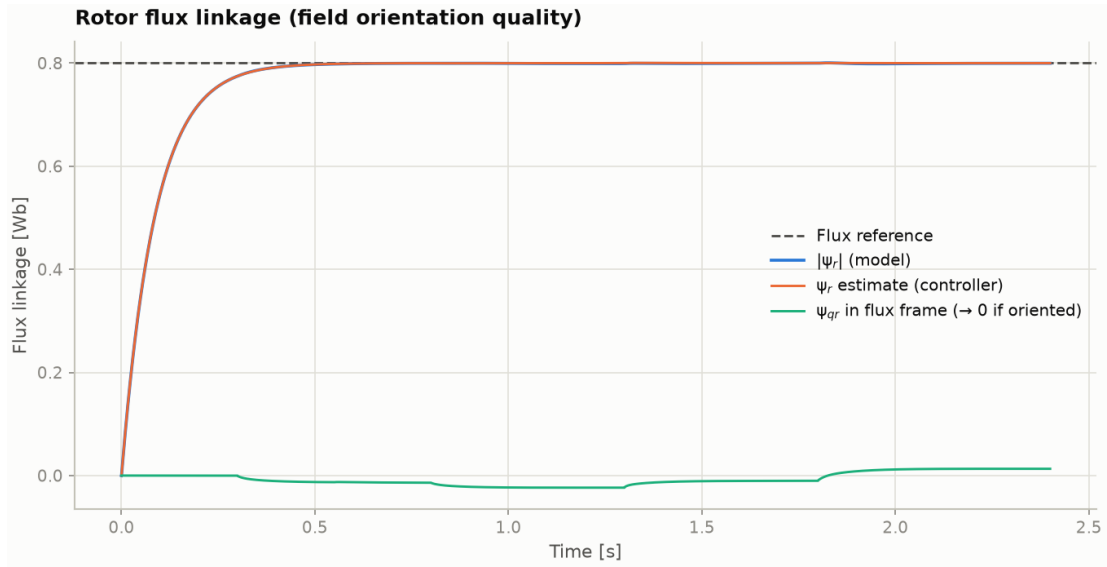


Figure 4. Rotor flux during the whole run: modelled magnitude, controller estimate, and the quadrature component seen from the controller’s frame.

Figure 4 shows the flux build-up and its behaviour thereafter. The magnitude rises with the expected first-order response of time constant $\tau_r = 87$ ms and is settled after roughly $3\tau_r \approx 0.26$ s, which is why the reference profile allows 0.30 s before the first speed step. From that point on the flux stays pinned to its 0.8 Wb reference for the remaining two seconds, through four speed steps, the engagement of the load and the reversal.

Two observations establish that the field orientation is correct. First, the controller’s estimate lies on top of the modelled flux magnitude, which validates the current-model estimator (16). Second, and more importantly, the quadrature flux component measured in the controller’s own frame never exceeds about 0.02 Wb, that is, 2.5 % of the rated value, and settles back to 0.013 Wb at the end of the run. A perfectly oriented drive would show exactly zero. The residual is the transient orientation error of the indirect method, which appears during fast changes of slip and decays afterwards. Its magnitude is the expected signature of a correctly functioning IFOC scheme; a value of several tens of percent would have indicated a detuned τ_r or a sign error in the transformations.

6.2 Speed tracking

Every step in Figure 5 produces the same signature: a torque-limited transient followed by an exponential approach to the new level, with no overshoot and no steady-state error. The settling times measured into a ± 10 rpm band are 79 ms for the step to 800 rpm, 75 ms for the step to 1400 rpm, 71 ms for the deceleration to 600 rpm and 95 ms for the reversal step, which spans 1400 rpm.

During the transients the response speed is set by the 35 N·m torque ceiling rather than by the loop gains. This is visible in the shape of the trace: the speed rises along a straight segment, corresponding to constant acceleration under constant torque, and the linear behaviour of the loop takes over only in the last part of the transient. That the settling times are nevertheless comparable across steps of very different amplitude is a direct consequence: a step twice as large spends roughly twice as long in saturation, but the final linear approach is identical.

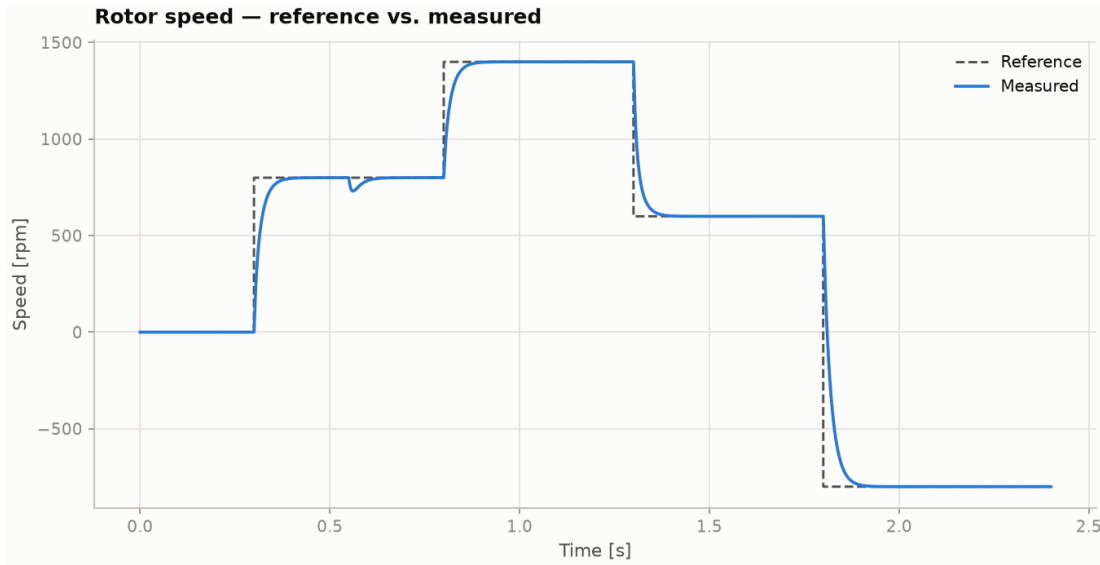


Figure 5. Speed response to the staircase of references, with the 10 N·m load step applied at $t = 0.55$ s.

The reversal at $t = 1.80$ s is handled without any dedicated logic. The drive brakes, passes through zero speed and re-accelerates in the opposite direction, which is the four-quadrant capability that vector control provides naturally. There is no discontinuity, no re-magnetisation and no loss of orientation at zero speed, because the flux angle is obtained by integration of $p\omega_m + \omega_{sl}$ and remains well defined even when ω_m crosses zero.

Between transients the error essentially vanishes: 0.05 rpm RMS with a maximum of 0.25 rpm over all settled portions. This is the internal-model principle at work, and Section 7.1 makes the argument explicit.

The load step at $t = 0.55$ s produces a dip of 69 rpm which is recovered in about 70 ms. This is the direct payoff of the 30 Hz speed-loop bandwidth chosen in Section 4.7, and it matches the design estimate given there.

6.3 Torque

The torque trace of Figure 6 reads as a direct commentary on the speed profile. Each upward speed step is driven by a saturated +35 N·m segment. At $t = 0.55$ s the torque steps to 10 N·m as the load engages. The deceleration from 1400 to 600 rpm and the reversal both produce saturated segments at −35 N·m: the machine brakes regeneratively, operating briefly as a generator while the kinetic energy stored in the shaft is returned to the DC bus through the inverter.

After each saturated segment the torque recovers cleanly, with no overshoot and no residual transient. This is the visible effect of the anti-windup: the integrators of the current and speed controllers do not charge during saturation, so nothing has to be discharged when the controller returns to its linear range.

On each plateau the torque settles on the load line $T_L + f_r\omega_m$. At −800 rpm this gives approximately 9.9 N·m, slightly less than the 10 N·m load, because the friction term is negative when the speed is negative and therefore assists the motor.

6.4 Currents in the rotor-flux frame

Figure 7 is the clearest demonstration of the decoupling. The flux-producing current i_{sd} jumps to its reference of 11.6 A at $t = 0$ and stays there for the entire run, regardless of what the q

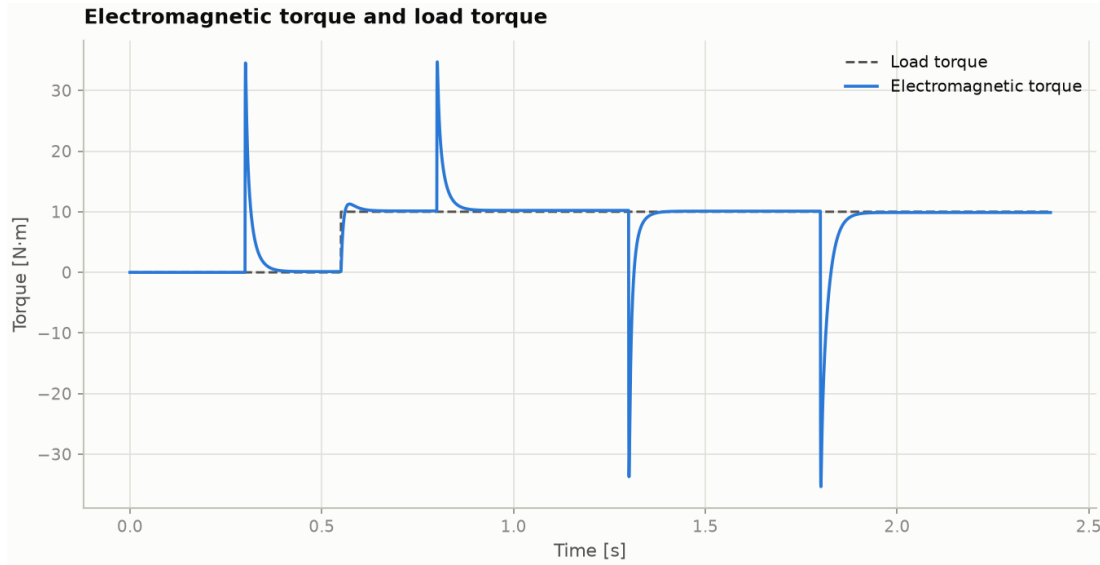


Figure 6. Electromagnetic torque during scenario A. The saturated segments at ± 35 N·m correspond to the torque limit imposed on the speed controller output.

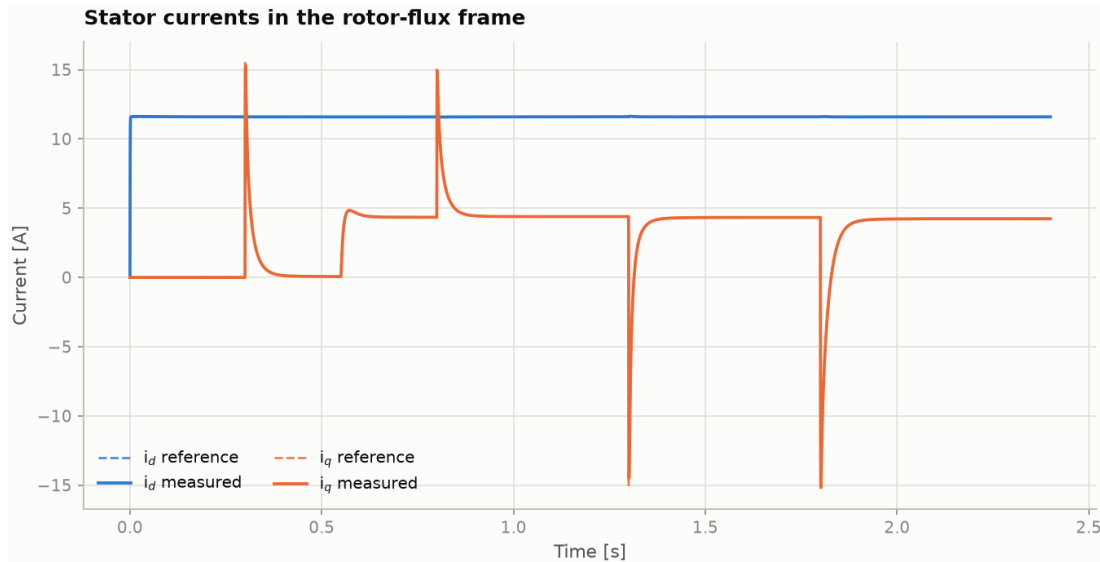


Figure 7. Flux- and torque-producing current components. References are drawn dashed; they are hidden behind the measured traces at this scale.

axis is doing: four speed steps, repeated torque saturation, the engagement of the load and a full reversal leave it unmoved. This is exactly the property that field orientation is supposed to deliver, and it is not obtained by construction but as a result of the decoupling terms (18) and of the correct flux angle.

The torque-producing current i_{sq} is a scaled copy of the torque trace, as expected from (14) at constant flux: spikes at +15 A during upward steps, −15 A during braking transitions, and a plateau near 4.3 A on the loaded segments. The value 15 A is simply the 35 N·m torque limit divided by $\frac{3}{2}p \frac{L_m}{L_r} \psi_r$.

Reference and measured traces are indistinguishable at the scale of the figure. The 400 Hz current loops track their references within the line width, which is why the dashed reference curves are invisible; this is the practical meaning of a time-scale separation of thirteen between

the loops.

6.5 Stator phase currents

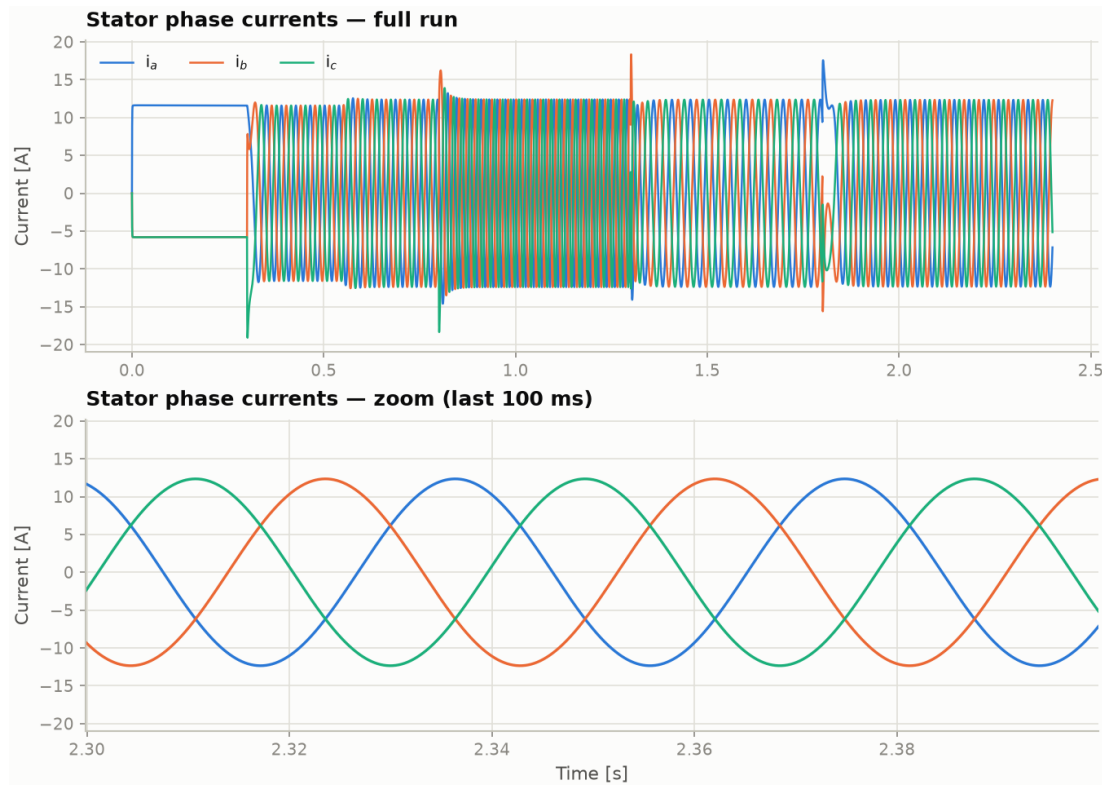


Figure 8. Stator phase currents over the whole run, with a zoom on the final steady state at -800 rpm.

Figure 8 translates the vector picture back into physical winding quantities, and it is instructive precisely because it looks so different.

During the magnetisation phase the phase currents are constant: $i_a = 11.6$ A and $i_b = i_c = -5.8$ A. This is not an artefact. The current space vector has a magnitude of 11.6 A and does not rotate, because the flux frame itself does not rotate when the machine is at standstill with no torque demand. A rotating field requires either rotation or slip, and at this point there is neither.

Each speed plateau then exhibits a distinct constant frequency, given by $f_e = (p \omega_m + \omega_{sl}) / 2\pi$, with short amplitude swells during the torque-limited transients and a permanent amplitude increase after the load engages. During the reversal the frequency sweeps down through a very low value as the electrical speed crosses zero, then increases again with the opposite phase sequence. The zoom shows the final steady state: three balanced sinusoids of about 12.3 A amplitude at approximately 27 Hz. The peak current over the whole run is 19.1 A, comfortably inside the 25 A limit, which confirms that the torque ceiling of 35 N·m was chosen consistently with the current limit.

6.6 Controller output voltages

The q -axis voltage of Figure 9 carries the back-EMF and therefore follows the speed levels: about 250 V at 1400 rpm, 110 V at 600 rpm, and a negative value at -800 rpm, with short excursions towards the limit during the step transients. The d -axis voltage remains small, close to $R_s i_{sd}$ plus the coupling correction.

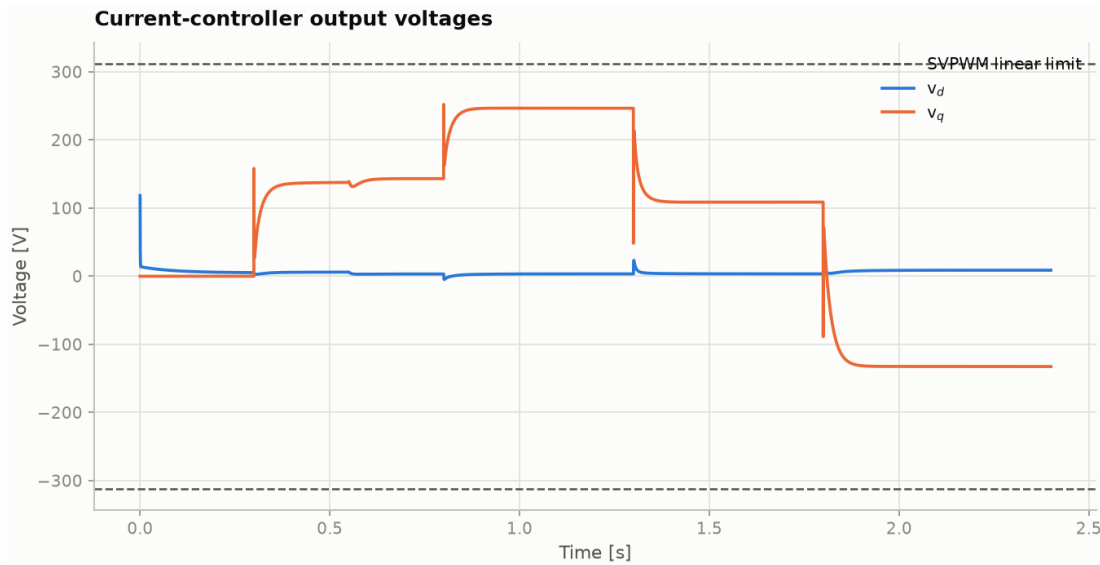


Figure 9. Voltage commands produced by the current controllers, with the linear SVPWM limit indicated.

The magnitude of the voltage vector stays inside the linear SVPWM limit of 311.8 V at all times. This confirms the design intent stated in Section 4.6: the operating envelope was chosen so that the drive never needs field weakening, which keeps the constant-flux assumption valid throughout and makes the interpretation of every other figure straightforward.

6.7 Summary of measured performance

Table 6. Scenario A: summary of measured quantities.

Metric	Value
Settling time, ± 10 rpm band (800 / 1400 / 600 / -800 rpm)	79 / 75 / 71 / 95 ms
Overshoot on any step	none
Speed error on settled plateaus, RMS	0.05 rpm
Speed error on settled plateaus, maximum	0.25 rpm
Load-step dip (10 N·m at 800 rpm) and recovery	69 rpm, ≈ 70 ms
Final speed error on the -800 rpm plateau	0.0 rpm
Rotor flux at end of run (reference 0.80 Wb)	0.800 Wb
Orientation residual $ \psi_{rq} $ at end of run	0.013 Wb (1.6 %)
Peak phase current (limit 25 A)	19.1 A

7 Results: scenario B, servo behaviour of the speed loop

7.1 What a PI loop can and cannot track

Zero steady-state tracking error requires the open loop to contain a model of the reference signal. This is the internal-model principle, and applying it to the speed loop explains the results of both scenarios at once.

The open-loop transfer function of the speed loop is approximately

$$L(s) = \underbrace{\left(K_p + \frac{K_i}{s}\right)}_{\text{PI}} \cdot \underbrace{\frac{1}{Js}}_{\text{mechanics}}, \quad (24)$$

neglecting the friction and treating the current loop as ideal within its bandwidth. It contains two integrators: one from the controller and one from the inertia. The consequences follow directly from the reference class.

Steps require one integrator in the open loop. The loop has two, so steps are tracked with zero steady-state error. This is what Section 6.2 observes on every plateau.

Ramps require two integrators. The loop is of type 2, so ramps are also tracked with zero steady-state error.

Sinusoids require a pair of resonant poles at the excitation frequency. The PI controller contains no such term, so a sinusoidal reference is tracked with a finite, frequency-dependent error. From $E(s) = R(s)/(1 + L(s))$, a sine of amplitude A and angular frequency ω leaves an error of amplitude

$$|E| \approx A \frac{J \omega^2}{|K_p J \omega + K_i|}, \quad (25)$$

which grows roughly as ω^2 at low frequency.

Equation (25) has a simple interpretation: the acceleration demand grows as ω^2 while the available loop gain grows only as ω , so the loop progressively runs out of gain as the reference becomes more demanding.

7.2 Measured tracking performance

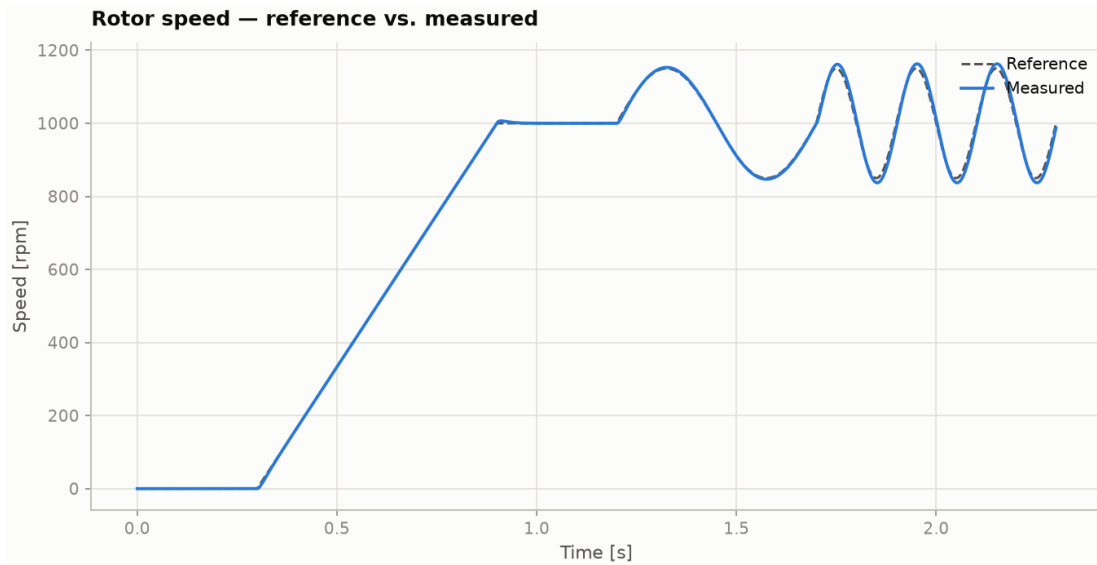


Figure 10. Scenario B: speed response to the ramp and to the two sinusoidal segments.

Table 7 compares the measured errors with the analytical predictions. The agreement is close enough to be conclusive.

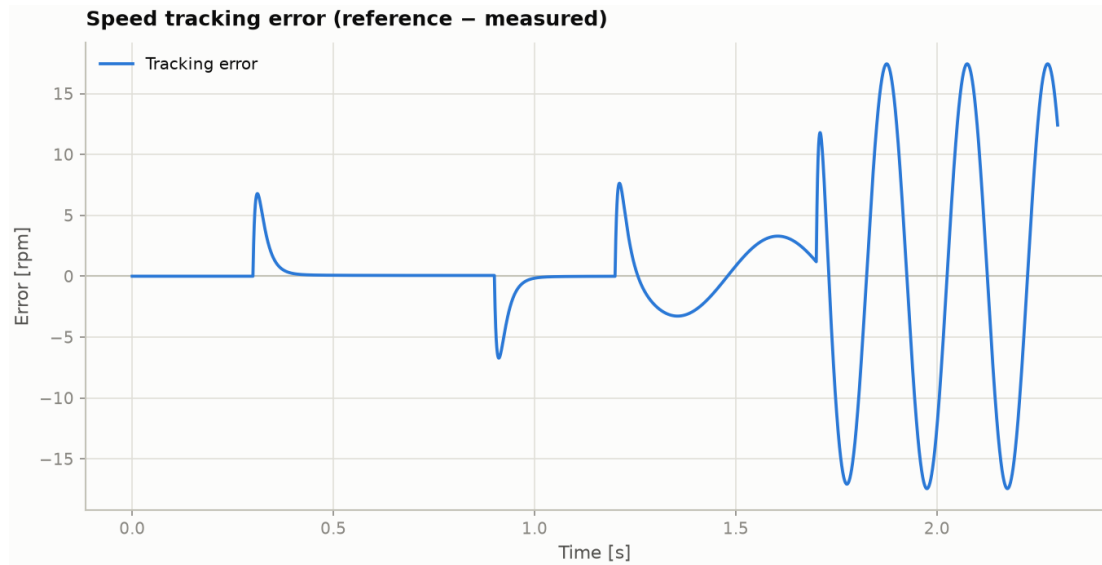


Figure 11. Scenario B: tracking error. The vertical scale makes visible what Figure 10 cannot show.

Table 7. Scenario B: measured versus predicted tracking error.

Segment	Measured max	Measured RMS	Predicted
Ramp, steady part	0.08 rpm	0.07 rpm	≈ 0 (type-2 loop)
Hold at 1000 rpm	0.16 rpm	0.04 rpm	0
Sine 2 Hz, ± 150 rpm	3.29 rpm	2.41 rpm	3.16 rpm (2.1 % of A)
Sine 5 Hz, ± 150 rpm	17.45 rpm	12.63 rpm	16.00 rpm (10.7 % of A)

The ramp is followed with essentially zero error once the short transient at the slope discontinuity has decayed. The small excursions of about ± 7 rpm visible at $t = 0.30$ s and $t = 0.90$ s in Figure 11 are the loop absorbing a sudden change in reference slope: the onset of a ramp is a step in the acceleration demand, and the loop responds to it as it would to any step disturbance.

The 2 Hz sinusoid leaves a barely visible error of 3.3 rpm. At 5 Hz the error grows to 17.5 rpm, a factor of 5.3 for a frequency ratio of 2.5, which is close to the ω^2 law predicted by (25). The residual discrepancy of roughly 9 % between prediction and measurement comes from two effects that the formula ignores: the viscous friction, which adds a small amount of natural damping, and the finite bandwidth of the current loop, which adds phase lag. The error is almost pure phase lag, as can be seen from its envelope, which is sinusoidal at the excitation frequency and in quadrature with the reference.

Figure 12 closes the argument. Throughout the whole servo exercise i_{sd} never departs from 11.6 A while i_{sq} supplies the small oscillating acceleration torque, which never exceeds a few newton-metres here. The flux/torque decoupling therefore holds during dynamic trajectory tracking exactly as it did during the step scenario, which was not guaranteed a priori: decoupling relies on feed-forward terms evaluated at the current operating point, and a rapidly varying operating point could in principle degrade it.

7.3 Practical reading

The practical conclusion for drive design is that a PI speed loop is the correct and sufficient tool when the application's references are set-points and its disturbances are load steps, which

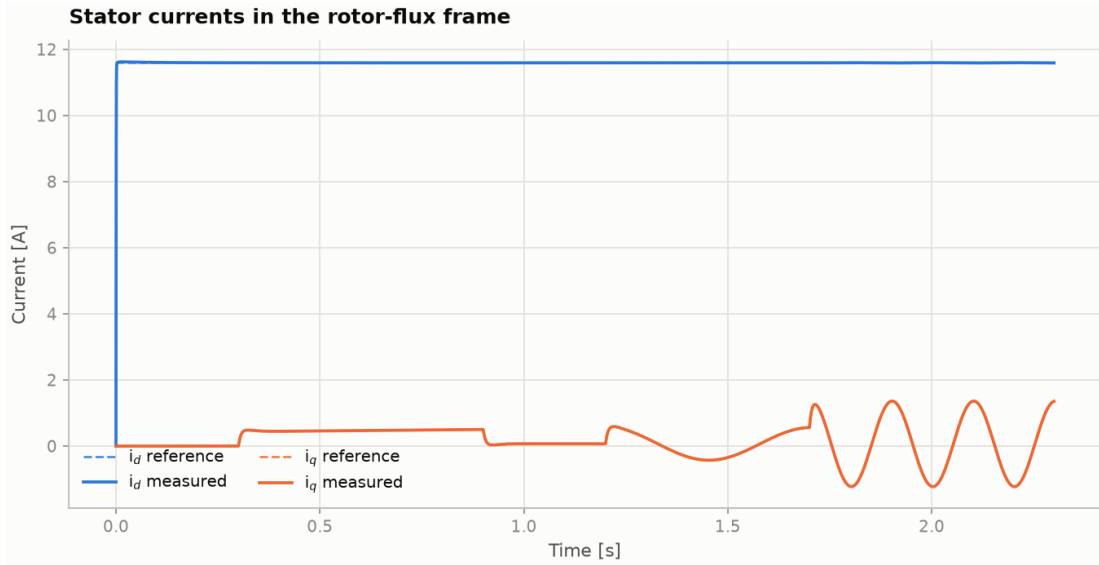


Figure 12. Scenario B: dq currents. The flux-producing component remains constant while the torque-producing component supplies the oscillating acceleration torque.

covers the overwhelming majority of industrial cases. When a drive must instead follow a trajectory, as in positioning axes or electronic cams, the appropriate response is to add the missing internal model, either through a proportional-resonant term at the relevant frequency or through a reference feed-forward path. Raising the PI gains is not an equivalent solution: it would trade tracking accuracy against noise sensitivity and robustness, and the improvement obtained would be proportional to ω while the requirement grows as ω^2 .

8 MATLAB and Simulink implementations

The drive was rebuilt twice in Simulink R2024b, first to cross-check the Python results on an independent platform and second to study the effect of a genuine switching inverter. Both versions use the machine data of Table 2, the gains of Table 3 and the scenario of Table 4. The tracking study of Section 7 is run only by the Python implementation.

Table 8. The two Simulink implementations.

	Version 1 (simulink_blocks)	Version 2 (simulink_library)
Built from	fundamental Simulink blocks only	Simscape Electrical components
Machine	hand-built dq equations	Asynchronous Machine, SI units
Inverter	average-value SVPWM	Universal Bridge (IGBT) + PWM generator
Controller	continuous	discrete
Solver	ode23tb, variable step	fixed step, discrete, $2.5 \mu\text{s}$
Run time (2.4 s of simulation)	a few seconds	2 to 3 minutes

Both models are parameterised entirely from a structure P built by the initialisation script `foc_init.m`. Every block dialog references a field of P rather than a hard-coded number, so the two Simulink versions and the Python simulation always run on the same machine data and the same gains. This is a small implementation detail with a large consequence: it removes the most common cause of spurious disagreement between models.

8.1 Version 1: built from fundamental blocks

The top level of Version 1 was already shown in Figure 1 and the controller in Figure 2, since both figures are exported from this model. Two subsystems deserve a closer look.

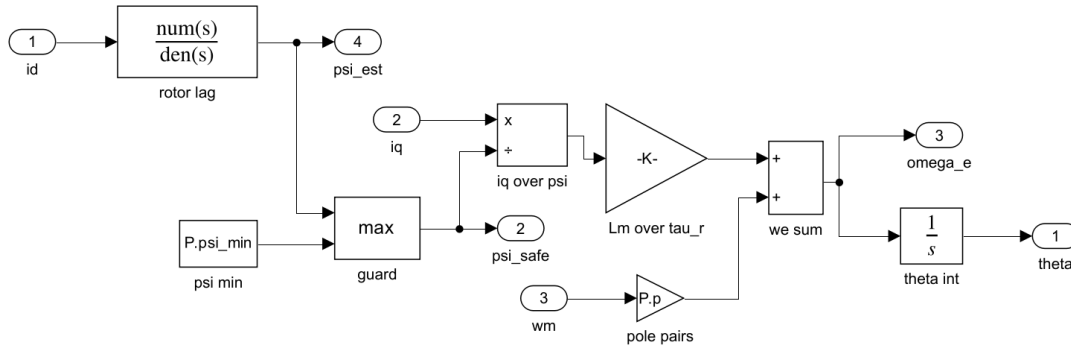


Figure 13. Flux estimator and angle generator: the rotor lag of Equation (16), the slip computation with its division guard, and the integrator producing θ_e .

The estimator of Figure 13 implements (16), guards the slip computation against division by zero at start-up when $\hat{\psi}_r \rightarrow 0$, forms ω_{sl} and integrates $\omega_e = p\omega_m + \omega_{sl}$ into the flux angle. The guard is essential rather than defensive: without it the very first sample after reset produces an infinite slip.

The machine model of Figure 14 is the state-space model of Section 2.3 drawn out block by block. Building it by hand rather than using a library component is what makes Version 1 an independent check: it shares no code with either the Python model or the Simscape model.

Measured effect of the anti-windup. Version 1 was used to quantify the anti-windup claim of Section 4.3. With the back-calculation loop of Figure 3 replaced by plain integrator clamping, the speed response to the 800 rpm step overshoots to approximately 960 rpm, that is, 20 %, and the settling time increases by roughly 20 %. With back-calculation active the overshoot disappears completely. The mechanism is straightforward: during the saturated phase the speed error remains large and positive, so the integral state continues to accumulate even though the output is already at its limit; when the speed finally reaches the reference, that accumulated state has to be discharged, which requires an error of the opposite sign, that is, an overshoot.

8.2 Version 2: built from library components

In Version 2 the power stage is physical. A DC voltage source feeds an IGBT Universal Bridge that switches at 10 kHz, current measurement blocks provide the feedback, and the machine is the Asynchronous Machine block of Simscape Electrical, with `powergui` configured in discrete mode at $2.5 \mu\text{s}$. The controller subsystem keeps the same structure but uses the library `abc` to `dq0` and `dq0` to `abc` transformation blocks and discrete PID blocks.

One implementation detail is worth recording, because it is invisible until the drive fails to magnetise and is then difficult to diagnose. The SPS transformation blocks are sine-referenced,

$$d_{\text{SPS}} = \frac{2}{3} \sum_k x_k \sin\left(\omega t - k \frac{2\pi}{3}\right), \quad (26)$$

whereas the Park transformation of Section 3.3 is cosine-referenced. The two are related by $q_{\text{SPS}} = d$ and $d_{\text{SPS}} = -q$, so the controller must contain an explicit axis swap and sign inversion

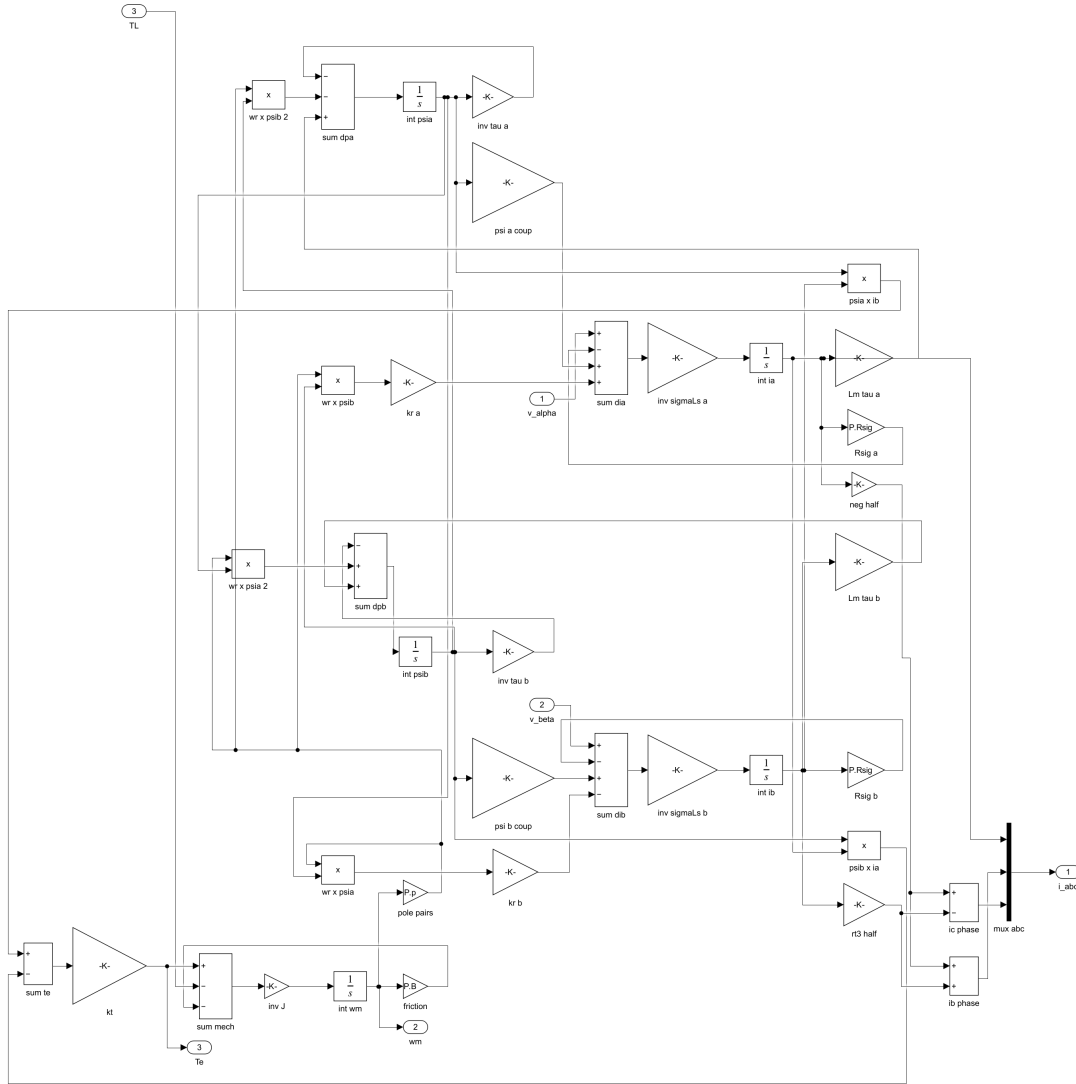


Figure 14. Machine model drawn explicitly: four integrators for the states of Equations (4)–(7), the cross-coupling products with ω_r , the torque equation and the mechanical integrator.

on both the measurement path and the actuation path. Without it the reference frame sits 90° away from the rotor flux, the current that was meant to produce flux produces torque instead, and field orientation is lost completely. The symptom is a machine that draws current, produces erratic torque and never magnetises, which can easily be mistaken for a tuning problem.

8.3 Results and three-way comparison

The switching model reproduces the same staircase response as the averaged models, with one visible difference: the phase currents carry the PWM ripple, and the speed trace shows a corresponding fine ripple on each plateau. Table 9 collects the quantitative comparison.

The three implementations agree on every physically meaningful quantity. The load-step dip is the most telling of these: it depends only on the machine inertia and the speed-loop bandwidth, not on how the inverter or the machine is represented, and the three values agree within 1 %. The rotor flux lands on its reference in all three.

The differences that remain are explained by what each model represents, and it is worth being explicit about them rather than treating them as noise.

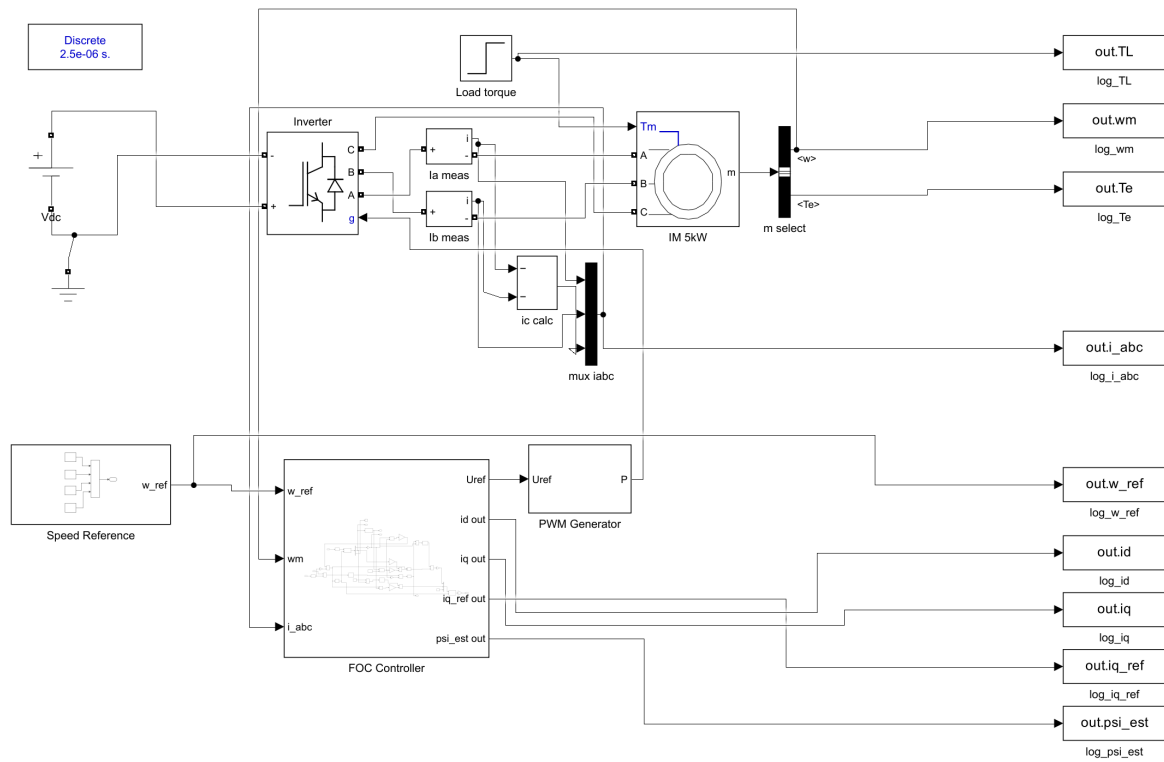


Figure 15. Version 2: DC source, IGBT Universal Bridge switching at 10 kHz, current measurements and the Asynchronous Machine block, solved by powergui in discrete mode.

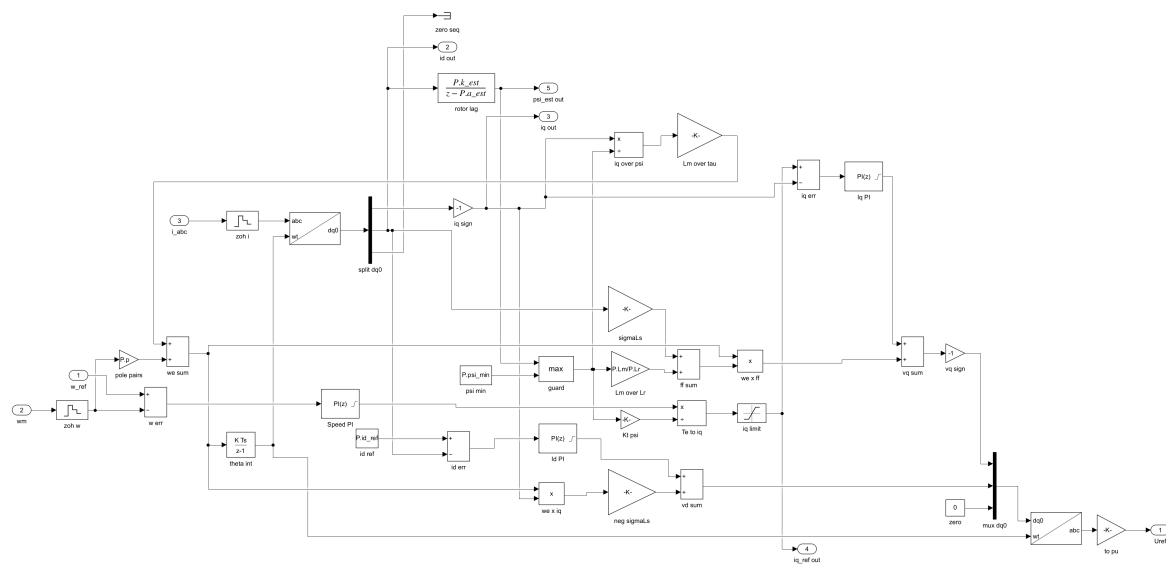


Figure 16. Version 2 controller. The axis swap and sign inversion required by the sine-referenced SPS transformation blocks are visible on both the measurement and the actuation path.

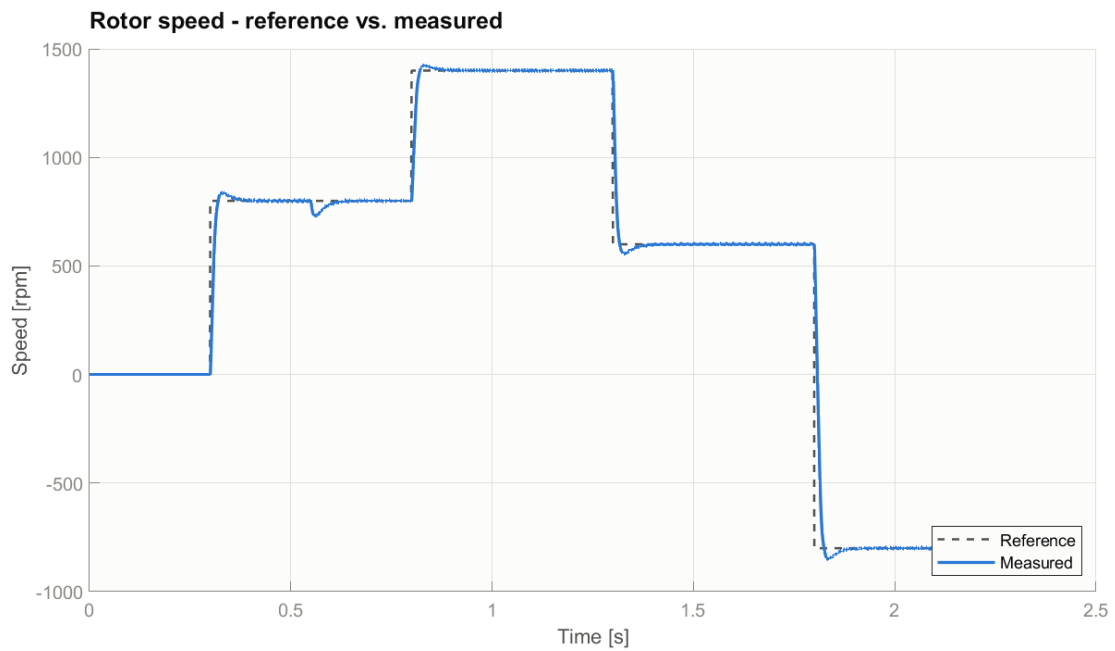


Figure 17. Version 2: speed response. The staircase is reproduced, with a fine ripple on each plateau caused by the switching inverter.

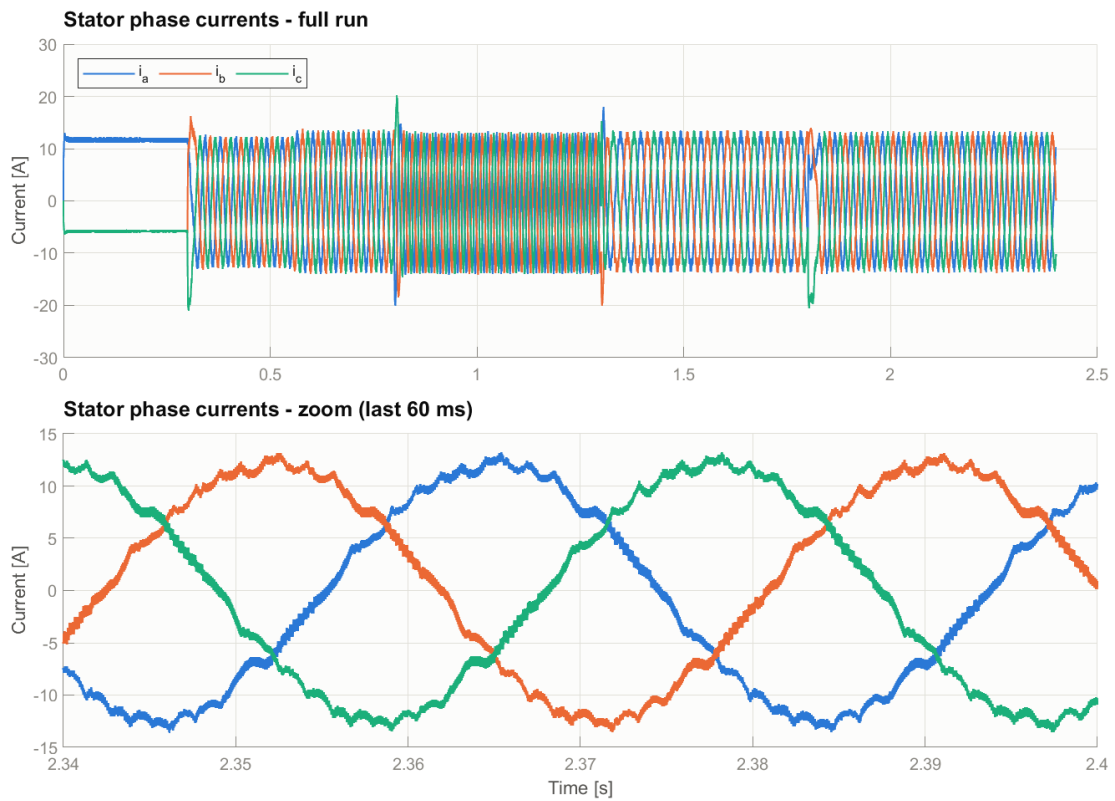


Figure 18. Version 2: stator phase currents, now carrying the PWM ripple that the averaged models smooth away.

Table 9. Three-way comparison of the implementations on scenario A.

Metric	Python	Simulink V1	Simulink V2
Settling times [ms]	79 / 75 / 71 / 95	70 / 68 / 52 / 59	63 / 62 / 67 / 72
Plateau error, maximum [rpm]	0.25	0.16	4.94
Load-step dip [rpm]	69.0	69.2	69.6
Rotor flux at end [Wb]	0.800	0.800	0.800
Peak phase current [A]	19.1	19.9	20.6

- The Python model runs a strictly discrete controller at $100\ \mu\text{s}$. The extra half-period of average delay slows the closed loop slightly, which is why its settling times are the longest of the three.
- Version 1 uses a continuous controller with an averaged inverter, so it has neither the sampling delay nor the switching ripple. It is therefore the fastest, and its settling times should be read as an upper bound on achievable performance rather than as a realistic prediction.
- Version 2 has a real switching bridge. Its plateau error of 4.94 rpm is an order of magnitude larger than in the averaged models, and its peak current is 8% higher. Both are direct consequences of the PWM ripple, which the averaged models discard by construction. This is precisely the quantity that the averaged model cannot predict and that motivates building the switching model at all.

That three independent implementations converge on the same numbers is the strongest available evidence that the machine model, the control design and the parameter set are all correct.

9 Discussion and limitations

The results presented above are simulation results, and the assumptions listed in Section 2 bound what can legitimately be concluded from them. Four limitations deserve explicit mention.

Parameter sensitivity. The indirect scheme relies on τ_r in both the slip relation (15) and the flux estimator (16). The rotor resistance of a cage machine varies by up to 50% between cold and hot operation, and the corresponding drift in τ_r produces a steady orientation error. The practical consequences are a loss of decoupling, so that i_{sd} no longer holds the flux alone, and an error in the torque constant, so that the commanded torque is no longer the produced torque. Neither effect causes instability, which is what makes them easy to overlook. Two remedies are standard: online adaptation of τ_r , typically from a model-reference scheme, or a switch to direct field orientation with a closed-loop flux observer, which replaces the open-loop dependence on τ_r by a feedback correction.

Ideal measurements. Current and speed measurements are assumed perfect. A real drive measures two phase currents with a finite-bandwidth sensor and an ADC, and obtains speed by differentiating an encoder position, which amplifies quantisation noise at exactly the frequencies where the speed loop is most sensitive. In practice this places a further upper bound on the speed-loop bandwidth, typically below the value used here, and often requires filtering

that adds phase lag.

Linear magnetic circuit. Saturation is neglected. Since the drive operates at constant rated flux and never enters field weakening, this assumption is reasonable within the envelope studied. It would not survive an extension to field weakening, where the operating point moves along the magnetisation curve and L_m becomes a function of the flux level.

Scope of the validation. The three implementations validate each other, not the machine. Agreement between models built from the same parameter set demonstrates that the equations, the transformations, the tuning and the implementations are mutually consistent. It says nothing about whether the parameters themselves describe a real motor. Experimental validation on a test bench remains necessary before any claim about physical performance.

10 Conclusion and outlook

A complete field-oriented drive was modelled, designed and validated in simulation for a 5 kW squirrel-cage induction machine. The two-stage coordinate transformation, Clarke to collapse the redundant three-phase description onto two orthogonal axes and Park to freeze the rotation at the rotor-flux angle, converts the coupled alternating-current machine into two independent direct-current-like channels. The indirect field-oriented structure built on top of it, comprising a slip-based angle generator, a current-model flux estimator and cascaded PI controllers with decoupling feed-forward and anti-windup, exploits that structure fully.

The simulated drive holds the rotor flux constant to within the thickness of the plotted line while the torque channel independently executes a staircase of speed steps that includes regenerative braking and a full four-quadrant reversal. Each step settles in under 100 ms with no overshoot and no steady-state error, the 10 N·m load disturbance is rejected with a 69 rpm dip recovered in about 70 ms, and every actuator limit behaves as designed. The dedicated tracking study completes the control picture by confirming the internal-model prediction quantitatively: the loop is of type 2 and therefore exact on steps and on ramps, while sinusoidal references produce a finite error growing as ω^2 , which only an added internal model or a feed-forward path could remove. Measured and predicted values agree within roughly 9 %.

The two Simulink implementations confirm the design independently. The fundamental-block model and the switching Simscape Electrical model reach the same settling times, the same 69 rpm load dip and the same rotor flux as the Python simulation, and the switching model additionally exhibits the PWM ripple that the averaged models omit by construction.

Three extensions follow naturally from this work. The most immediate is field weakening, which would extend operation above base speed and is now practicable since the switching behaviour has been validated. The second is an online adaptation of the rotor time constant, or a switch to a closed-loop flux observer, which would address the principal robustness weakness identified in Section 9. The third is a sensorless variant, replacing the measured speed by an observer and removing the encoder, which is the configuration most often required in industrial practice. Beyond these, moving the tuning from fixed bandwidth targets to an explicitly robustness-based design would treat the parameter sensitivity as a design constraint rather than as a limitation to be noted after the fact.

References

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A Derived parameters and controller constants

Table 10 collects every derived quantity used in the report, so that the numerical results can be reproduced without re-deriving them.

Table 10. Derived quantities and controller constants.

Quantity	Expression	Value	Unit
Leakage coefficient	$1 - L_m^2 / (L_s L_r)$	0.0555	–
Transient inductance	σL_s	3.94	mH
Transient resistance	$R_s + R_r (L_m / L_r)^2$	1.206	Ω
Rotor time constant	L_r / R_r	87.0	ms
Stator time constant	L_s / R_s	163.2	ms
Transient time constant	$\sigma L_s / R_\sigma$	3.27	ms
Torque constant	$\frac{3}{2} p L_m / L_r$	2.915	N·m/(Wb·A)
Flux reference	ψ_r^*	0.80	Wb
Flux current reference	ψ_r^* / L_m	11.6	A
Torque limit	$T_{e,\max}$	35	N·m
Current limit (peak)	$I_{\max}$	25	A
q -axis current limit	$\sqrt{I_{\max}^2 - i_{sd}^{*2}}$	22.1	A
DC-link voltage	V_{dc}	540	V
Linear SVPWM voltage limit	$V_{dc} / \sqrt{3}$	311.8	V
Current-loop bandwidth	$\omega_{ci} / 2\pi$	400	Hz
Speed-loop bandwidth	$\omega_{cs} / 2\pi$	30	Hz
Sampling period	T_s	100	μ s
Integration step	h	20	μ s

B Discrete PI controller with anti-windup

All controllers in the drive use the same discrete implementation, given below in pseudocode. The back-calculation gain is set to $K_b = 1/T_s$, which unwinds the integrator within approximately one sampling period once the output leaves saturation.

```

1 def pi_step(ref, meas, state, Kp, Ki, Kb, Ts, u_min, u_max, ff=0.0):
2     error = ref - meas
3     u_unsat = Kp * error + state.integral + ff # unsaturated command
4     u = clamp(u_unsat, u_min, u_max) # applied command
5     # integrator update, corrected by the saturation error
6     state.integral += Ts * (Ki * error + Kb * (u - u_unsat))
7     return u

```

Listing 1. Discrete PI controller with back-calculation anti-windup, executed once per sampling period.

Two points are worth noting. The feed-forward term ff is added before the saturation, so that the anti-windup accounts for it; adding it after saturation would allow the controller to command a voltage outside the achievable range without the integrator ever being informed. The integral state is updated after the saturation is known, which is what allows the correction term $K_b(u - u_{\text{unsat}})$ to be computed in the same pass.

C Repository organisation and reproduction

The code accompanying this report is organised as follows.

Table 11. Organisation of the accompanying code.

Path	Contents
python/	machine model, SVPWM inverter, controllers, scenarios, plotting
python/outputs/graphs/	every figure of Sections 6 and 7
matlab/m_files/	foc_init.m (parameter structure P), plotting and metrics
matlab/simulink_blocks/	Version 1, built from fundamental blocks
matlab/simulink_library/	Version 2, Simscape Electrical with switching bridge
matlab/outputs/	exported diagrams and result graphs

To reproduce the results, run the Python entry point to regenerate scenarios A and B and all associated figures, then execute `foc_init.m` in MATLAB to populate the parameter structure and run either Simulink model. The Simulink models read their parameters exclusively from that structure, so no value needs to be edited inside a block dialog.